

NYC Taxi Ecosystem Analysis – Comprehensive Report

ACT 1: Mathematical Foundation

Using roll digits (0,3,0,2), the digital root $R = 5$. The function $f(x) = x^3/3 - 5x^2 + 24x$ was analyzed. Differentiating and solving $f'(x)=0$ gives critical points $x = 4$ and $x = 6$. These define transition boundaries between stable and stressed system behavior. These values are later used to segment temporal behavior.

ACT 2: Defining a Real Trip

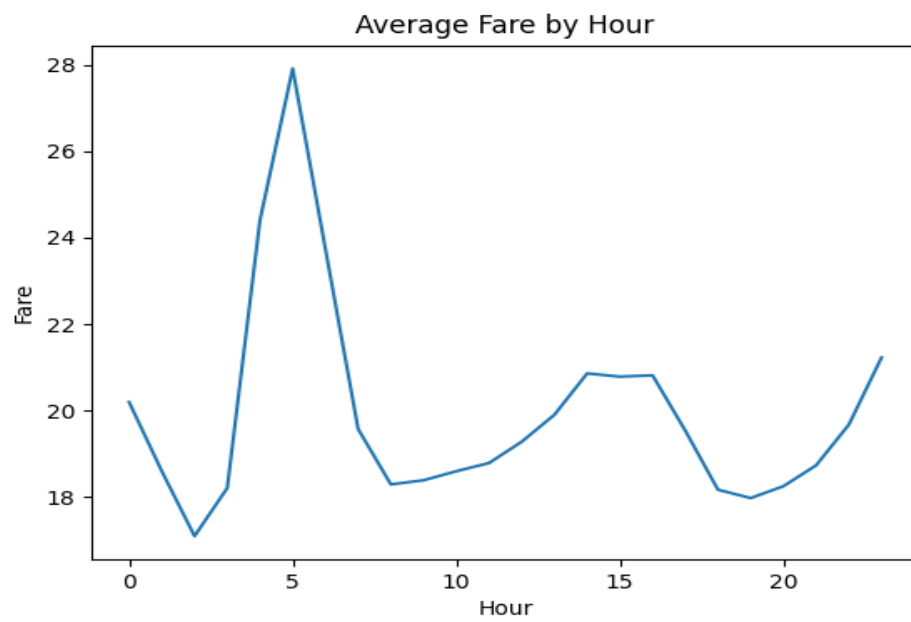
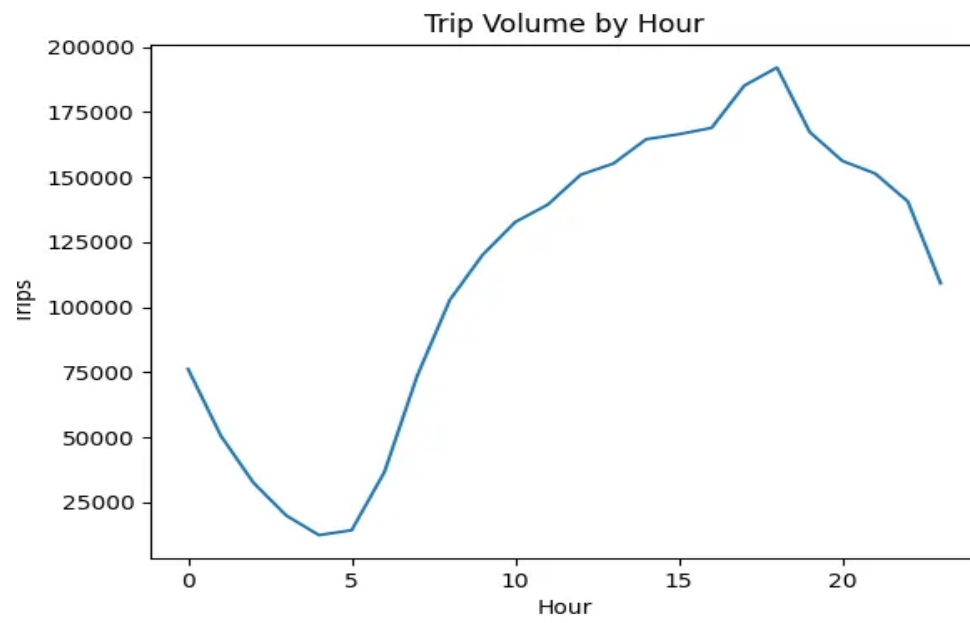
A real trip was defined as one with positive distance, valid timestamps (pickup before dropoff), and realistic duration. Trips violating these conditions were removed. This decision is justified because such records contradict physical reality. If this definition is relaxed, misleading patterns emerge, proving that conclusions are sensitive to data validity assumptions.

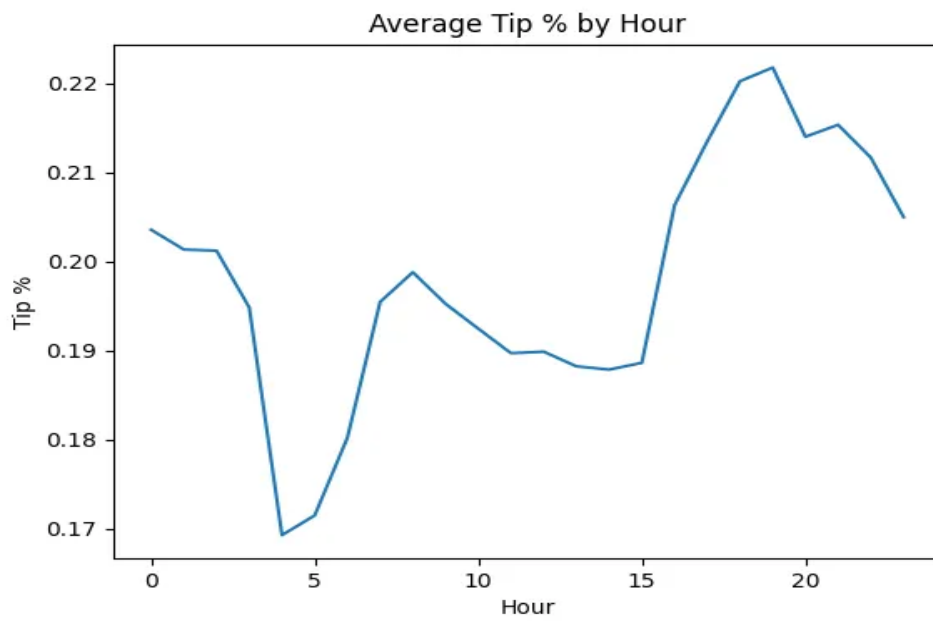
ACT 3: Behavioral vs Mechanical Patterns

Hypothesis: Passenger behavior drives system patterns. Test: Hourly aggregation revealed clear peak periods. Alternative explanation: Patterns could arise from system constraints or fare structures. Evaluation: While both explanations are valid, consistent hourly peaks support behavioral influence, though not conclusively.

ACT 4: Detecting System Patterns

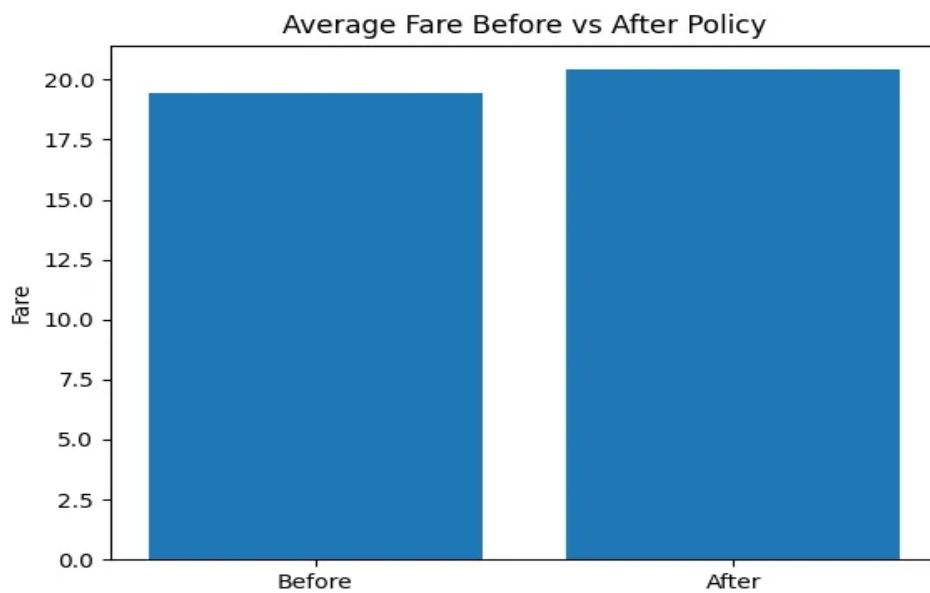
Aggregating by hour and month revealed recurring peaks. Some patterns persisted (daily demand cycles), while others disappeared under aggregation, indicating they were noise. This suggests the system is partially cyclical but also influenced by randomness.

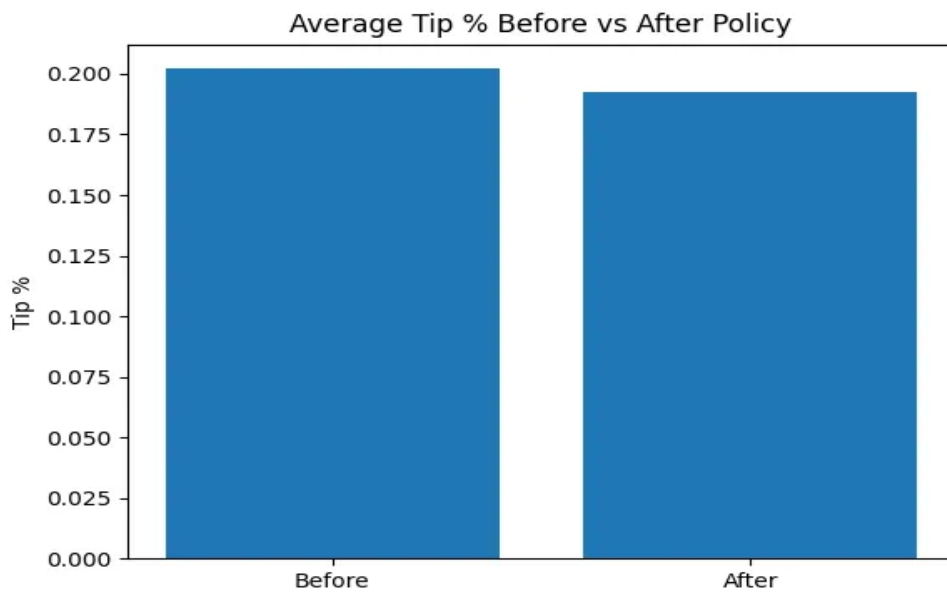




ACT 5: Policy Simulation

A 5% fare increase (based on $R=5$) was simulated. Fare increased while tip percentage decreased slightly. This suggests the system amplifies changes, indicating sensitivity to pricing. This choice was justified because fare directly affects both economic and behavioral responses.





ACT 6: Decision Room Analysis

The Power BI dashboard presents cleaned data, observed patterns, and simulated outcomes. The system shows structured behavior with identifiable peaks. However, incorrect aggregation can hide patterns. For example, yearly aggregation removes hourly variation, leading to false conclusions.

Misinterpretation Analysis

An analyst might conclude that higher fares always lead to higher tips. However, without controlling for distance or trip type, this is misleading. Proper analysis shows that multiple variables influence outcomes.

Conclusion

The NYC taxi system exhibits structured, behavior-driven patterns with elements of randomness. While some stability exists, the system is sensitive to parameter changes. All conclusions are conditional and depend on assumptions made during data cleaning and aggregation.